

L2: Basics of Linear Algebra

Topics to be covered today:

- Vectors
- Inner product
- Linear independence
- Basis, dimension of a subspace
- Matrices
- Rank of a matrix

Def: A vector is an ordered finite list of numbers.

$$\text{Eg: } a = \begin{pmatrix} -1.1 \\ 2.3 \\ 4 \\ 7 \end{pmatrix}$$

↳ 4-dimensional vector

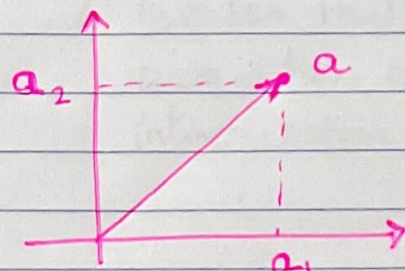
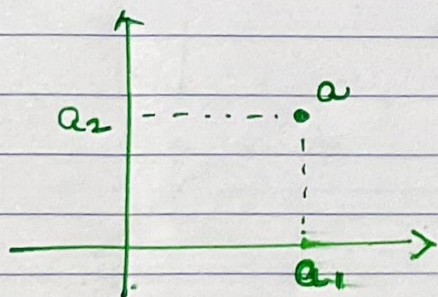
We say $a \in \mathbb{R}^4$ or that a is a vector of length/dimension 4.

a_i - i^{th} entry / component of a .

$$\text{Eg: } a_3 = 4$$

It is convenient to ~~visu~~ visualize a vector as:

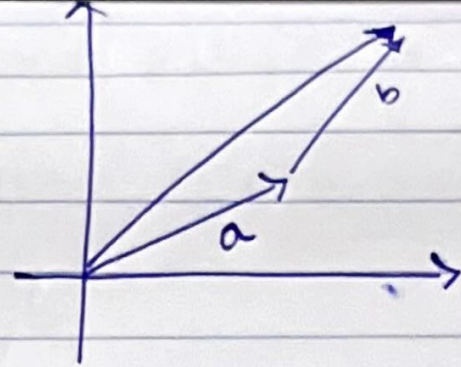
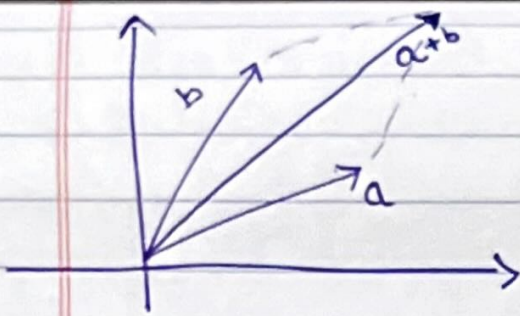
- a point in $\mathbb{R}^n$
- an arrow from the origin to this point in $\mathbb{R}^n$



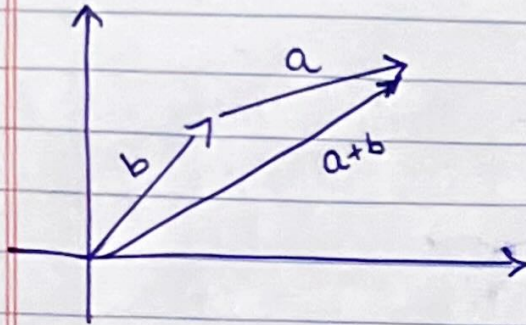
Vector addition

Vectors of the same size can be added (elementwise).

The visual interpretation of vector addition is useful.

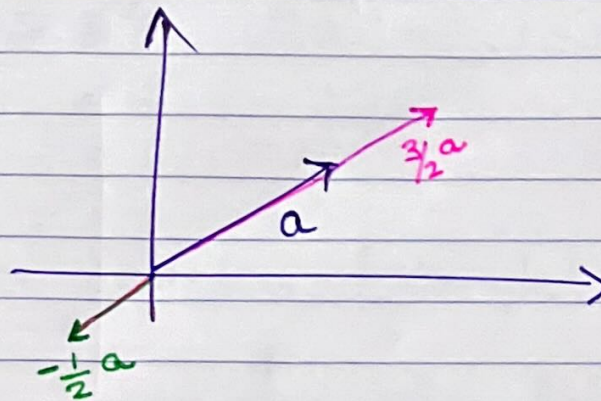


These visualizations are equivalent.



Scalar multiplication:

$$\underset{\substack{\downarrow \\ \text{scalar}}}{\alpha} \underset{\substack{\downarrow \\ \text{vector}}}{a} = \begin{pmatrix} \alpha a_1 \\ \vdots \\ \alpha a_n \end{pmatrix}$$



Scalar multiplication also has a useful interpretation

Inner / Dot product

Say $a, b \in \mathbb{R}^n$

$$\text{Then } \langle a, b \rangle = \sum_{i=1}^n a_i b_i$$

Also convenient to view inner product as a matrix multiplication:

View a & b as $n \times 1$ column matrices.

Then $\langle a, b \rangle = a^T b$

$$a = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

$$b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

$$a^T b = (a_1, \dots, a_n) \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a_1 b_1 + \dots + a_n b_n$$

The norm of a vector a is defined as:

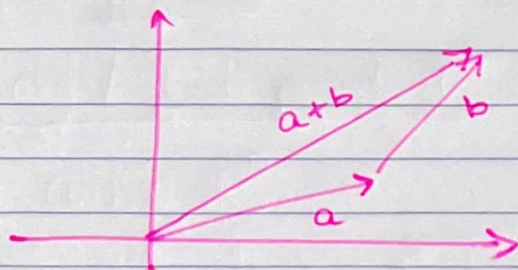
$$\|a\| = \sqrt{\langle a, a \rangle}$$

↓
magnitude of vector a

★ Important property of norms

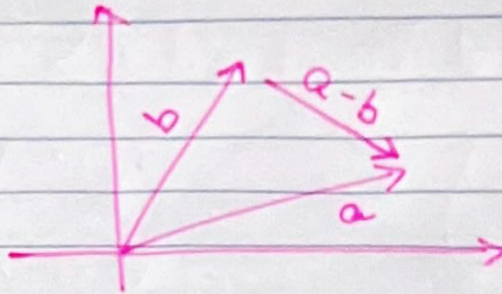
Norm satisfies triangle inequality:

$$\|a+b\| \leq \|a\| + \|b\|$$



The distance between vectors a & b is given by:

$$\text{dist}(a, b) = \|a - b\|$$



★ Another important property of inner product is norm

Cauchy - Schwartz inequality :

$$|\langle a, b \rangle| \leq \|a\| \cdot \|b\|$$



$$-\|a\| \cdot \|b\| \leq \langle a, b \rangle \leq \|a\| \cdot \|b\|$$

- You get equality on the right,
i.e. $\langle a, b \rangle = \|a\| \cdot \|b\|$

iff $a = \alpha b$, where $\alpha > 0$.

(Vectors are pointed in the same "direction")

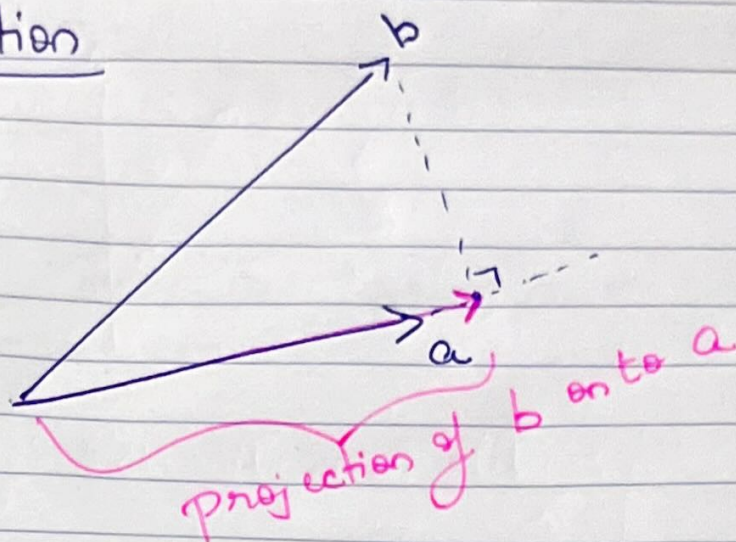
- You get equality on the left,
i.e. $\langle a, b \rangle = -\|a\| \cdot \|b\|$

iff $a = -\alpha b$, where $\alpha > 0$.

(Vectors are pointed in opposite "directions")

We say a is orthogonal / perpendicular to b if $\langle a, b \rangle = 0$.

Projection



The projection of b onto a is $\frac{\langle a, b \rangle}{\|a\|^2} a$

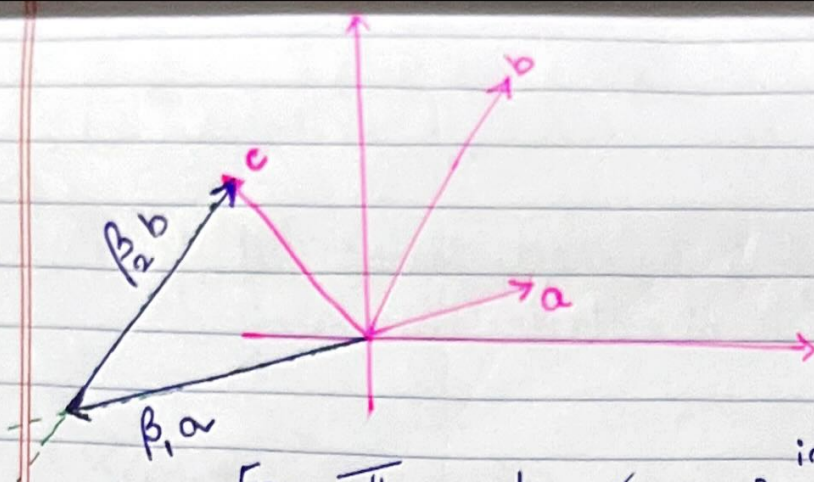
Magnitude of this projection is : $\frac{|\langle a, b \rangle|}{\|a\|}$.

Def A collection of vectors $\{a_1, \dots, a_k\}$ is linearly dependent if

$$\beta_1 a_1 + \beta_2 a_2 + \dots + \beta_k a_k = 0$$

holds for some $\beta_1, \dots, \beta_k$ that are NOT all zero.

{ This means at least one vector in the collection can be expressed as a linear combination of the others! }



Eg: The vectors $\{a, b, c\}$ ^{in $\mathbb{R}^2$} are ~~not~~ linearly dependent - We can write

$$c = \beta_1 a + \beta_2 b$$

$$\Rightarrow \beta_1 a + \beta_2 b + (-1)c = 0.$$

Def A collection $\{a_1, \dots, a_n\}$ is linearly independent if it is NOT linearly dependent.

{ This means no vector in this collection can be expressed as a linear combination of the others. }

Eg The collection $\{a, b\}$ in the above example is linearly independent.

Eg The collection $\{e_1, \dots, e_n\}$ of vectors in $\mathbb{R}^n$, where

unit $\leftarrow$ vectors $e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}$ $\leftarrow$ in position i

is linearly independent. - Exercise

Fact: If $\{a_1, \dots, a_k\}$ is an independent collection of vectors in $\mathbb{R}^n$, then $k \leq n$.

{ You cannot have a collection of 4 independent vectors in $\mathbb{R}^3$. }

Basis & dimension

Def A collection of n independent vectors in $\mathbb{R}^n$ forms a basis of $\mathbb{R}^n$.

Fact: If $\{a_1, \dots, a_n\}$ is a basis of $\mathbb{R}^n$, any vector $b \in \mathbb{R}^n$ can be expressed as a unique linear combination of the basis vectors,

$$\text{i.e., } b = \sum_{i=1}^n \beta_i a_i.$$

Eg $\{e_1, \dots, e_n\}$ just defined is a basis of $\mathbb{R}^n$.

In fact, this is an orthogonal basis.

{ any pair of basis vectors }
is orthogonal.

In fact, this is an orthonormal basis

{ orthogonal basis with
Unit norm basis vectors }

More generally, we can also ~~design~~ define basis & dimension of subspaces of $\mathbb{R}^n$.

Consider $\{a_1, \dots, a_k\}$.

Let L denote the set of all vectors that may be expressed as linear combinations of this collection.
→ subspace spanned by $\{a_1, \dots, a_k\}$

- If $\{a_1, \dots, a_k\}$ is linearly independent, it forms a basis of the subspace L .

We say L is k -dimensional.

- If not, there is a subset $\{a_{i_1}, \dots, a_{i_m}\}$ of the collection s.t. every vector in L is a unique linear combination of the vectors in this subset.

We say L is m -dimensional in this case.

Eg $\left\{ \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}, \begin{pmatrix} -3 \\ 4 \\ -1 \end{pmatrix} \right\}$
Span a 2-dimensional subspace of $\mathbb{R}^3$.

Exercise: Check that any two vectors of this collection are linearly independent, but

$$\begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix} = -\frac{1}{5} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} - \frac{7}{5} \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}.$$

Matrices

A matrix is a rectangular array of numbers.

Eg

$$A = \begin{bmatrix} 0 & 1 & -2.3 & 0.1 \\ 1.3 & 4 & -0.1 & 0 \\ 4.1 & -1 & 0 & 1.7 \end{bmatrix}$$

We say A is $m \times n$ if it has m rows & n columns.

Eg: Above matrix is 3×4 .

$A_{i,j}$

We use $A_{i,j}$ to denote the entry in i^{th} row & j^{th} col.

$A_{3,2}$

Eg $A_{3,2} = -1.$

$$\left\{ \begin{array}{l} m = n \Rightarrow \text{square matrix} \\ m > n \Rightarrow \text{tall matrix} \\ m < n \Rightarrow \text{wide matrix} \end{array} \right\}$$

For $m \times n$ matrix,

- There are m row vectors, each in $\mathbb{R}^n$

Subspace spanned by rows - row space

Dimension of row space - row rank

Note: $\text{Row rank}(A) \leq \min(m, n)$
of independent rows

- There are n column vectors, each in $\mathbb{R}^m$

Subspace spanned by columns - column space

Dimension of column space - column rank

Note: $\text{Col. rank of } A \leq \min(m, n)$
of independent columns

Facts

- $\text{Col. rank}(A) = \text{Row rank}(A) = \text{rank}(A)$
- ~~For~~ For square $n \times n$ matrix A ,

$\text{rank}(A) = n \iff A$ is invertible
 A is "full rank"

$\exists n \times n$ matrix A^{-1} s.t.
 $AA^{-1} = A^{-1}A = I$
identity matrix

Q: Can a tall matrix have full row rank?

Q: Can a tall matrix have full col. rank?